

Quantitative Well-Posedness of Mean-Field Games with Common Noise

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Abstract

We study mean-field games (MFGs) in which a continuum of agents interacts through a common, shared Brownian motion in addition to idiosyncratic noise. The equilibrium is characterised by a system of forward-backward stochastic differential equations (FBSDEs) coupled through the conditional law of the representative agent's state given the common-noise filtration. Building on the well-posedness theory of [Carmona and Delarue \(2018\)](#) and [Ahuja et al. \(2019\)](#), we derive an explicit, parameter-dependent contraction estimate for the associated fixed-point map in a weighted Wasserstein-2 metric. The contraction constant depends explicitly on the Lipschitz constant of the drift, the Frobenius norm of the common-noise coefficient, and bounds on the relevant Lions derivatives, and is computable from the model primitives. Existence and uniqueness of the mean-field equilibrium follow via the Banach fixed-point theorem, and we verify the associated ε -Nash property for the finite-player game. To the best of our knowledge, an explicit, computable contraction bound of this form – suitable as a design criterion for numerical algorithms – has not previously been derived in the common-noise setting; existing work establishes existence and uniqueness but not quantitative stability. We illustrate the theory on the canonical linear-quadratic MFG, for which all constants are explicit.

1 Introduction

Mean-field games (MFGs), introduced independently by [Lasry and Lions \(2007\)](#) and [Huang et al. \(2006\)](#), model the equilibrium behaviour of a large population of interacting agents, each optimising an individual objective that depends on the aggregate behaviour of the population. In the classical formulation, agents are subject only to idiosyncratic noise, and the mean-field limit is deterministic: the empirical measure converges to a deterministic flow of measures $(\mu_t)_{t \in [0, T]}$.

Many applications, however, involve a source of randomness shared by *all* agents simultaneously – a common macroeconomic shock, a shared environmental factor, or systemic market risk. This motivates the study of *common-noise* MFGs, in which each agent's state is driven by both an idiosyncratic Brownian motion and a shared Brownian motion W^0 . In this setting the equilibrium measure flow $\mu_t = \mathcal{L}(X_t \mid \mathcal{F}_t^{W^0})$ is itself a *stochastic* process, adapted to the common-noise filtration – a genuinely more delicate object than in the classical case.

The probabilistic well-posedness theory for common-noise MFGs was developed by [Carmona et al. \(2013\)](#) and treated comprehensively by [Carmona and Delarue \(2018\)](#), and extended under weaker (displacement) monotonicity conditions by [Ahuja et al. \(2019\)](#). These works establish existence and uniqueness of the equilibrium under monotonicity-type structural conditions, but do not provide *explicit, parameter-dependent* contraction constants: quantities that tell a practitioner, in terms of the model's Lipschitz constants and noise intensities, precisely how fast (and under what conditions) a fixed-point iteration converges.

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Contribution. We derive such an explicit contraction estimate. Working under standard Lipschitz, monotonicity, and regularity assumptions (Section 2), we show that the natural fixed-point map Φ associated with the common-noise MFG equilibrium satisfies a Wasserstein stability estimate (Theorem 3.1) whose constants are given explicitly in terms of the drift Lipschitz constant L_b , the common-noise coefficient norm $\|\sigma_0\|_F$, and bounds on the relevant Lions derivatives. This yields a contraction in a weighted Wasserstein metric (Theorem 4.1) with an explicit rate $\rho < 1$, from which existence, uniqueness, and the ε -Nash property for the associated N -player game follow. We illustrate all constants explicitly for the canonical linear-quadratic MFG.

We emphasise precisely what this contributes relative to the closely related recent work of Jackson and Tangpi (2024) and Jackson and Mészáros (2025), discussed in detail below: those papers establish quantitative *convergence rates of N -player Nash equilibria to an assumed-to-exist mean-field equilibrium* via displacement monotonicity, whereas the present paper constructs the mean-field equilibrium itself as the fixed point of Φ , via an explicit contraction estimate intended as a design criterion for a numerical fixed-point iteration scheme. To the best of our knowledge, relative to the specific works cited in this section, an explicit computable Banach-contraction estimate of this form for the equilibrium fixed-point map itself has not previously been derived in the common-noise setting. This scope reflects the mean-field-game literature most directly relevant to the present setting, including the papers identified in the focused literature search underlying this claim, and is not a claim of an exhaustive survey of all adjacent fields.

Related work. Cardaliaguet et al. (2019) derive the master equation for common-noise MFGs and establish PDE-level regularity; Cardaliaguet et al. (2025) resolve further SPDE regularity questions in this setting. Neither addresses quantitative FBSDE contraction rates for the fixed-point map Φ . For classical (idiosyncratic-only) MFGs, Cardaliaguet (2010) gives a PDE treatment of well-posedness; numerical schemes for this setting are studied by Achdou and Capuzzo-Dolcetta (2010) and Carmona and Laurière (2022). The propagation-of-chaos theory underlying the ε -Nash verification traces to McKean (1966) and Sznitman (1991), with quantitative Wasserstein rates due to Fournier and Guillin (2015).

Closely related work using synchronous coupling for common-noise FBSDE stability. Jackson and Tangpi (2024) use the same synchronous-coupling strategy underlying Section 3 to derive quantitative *convergence rates* $O(1/N)$ of N -player open-loop Nash equilibria to a mean-field equilibrium, for common-noise MFGs with controlled volatility, under a displacement-monotonicity structural condition. This replaces the small-time-horizon or dissipativity restriction used in earlier coupling-based arguments (e.g. the idiosyncratic-noise result of Laurière and Tangpi that they build on) with a monotonicity condition that holds more broadly. Jackson and Mészáros (2025) extend this to mean-field games of control. Both works also establish existence and uniqueness of the mean-field equilibrium, but via the *method of continuation* for monotone FBSDEs (in the sense of Peng and Wu, 1999), not via a Banach-fixed-point contraction estimate on the space of measure flows. The present paper’s target object – an explicit contraction constant for the map $\Phi : \mu \mapsto \Phi(\mu)$ itself, usable directly as a step-size / convergence-rate criterion for a numerical fixed-point iteration – is accordingly a different (though methodologically related) result. We view the displacement-monotonicity condition used by Jackson and Tangpi (2024); Jackson and Mészáros (2025) as a promising route to relaxing the smallness condition (16) required to close the Gronwall argument in Appendix A, and we flag this as a direction for future work. Finally, Friz et al. (2026) study a linear-quadratic MFG with *rough* (rather than Itô/Brownian) common noise and derive stability estimates with respect to initial data and the common-noise path in that pathwise setting; the technical setting (rough-path theory) and the notion of stability differ from the present Brownian, expectation-level analysis.

Organisation. Section 2 states the model, assumptions, and the fixed-point formulation. Section 3 contains the core stability estimate. Section 4 states and proves the main theorem (contraction, existence, uniqueness, ε -Nash). Section 5 discusses the contraction constant, its dependence on model parameters, and limitations of the current approach. Section 6 illustrates the theory on the linear-quadratic benchmark.

2 Model and Standing Assumptions

We work on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0, T]}, \mathbb{P})$ supporting an idiosyncratic Brownian motion W and an independent common Brownian motion W^0 . Let $\mathcal{P}_2(\mathbb{R}^d)$ denote the space of Borel probability measures on $\mathbb{R}^d$ with finite second moment, equipped with the Wasserstein-2 distance W_2 . The representative agent's state solves the McKean–Vlasov SDE

$$dX_t^\mu = b(t, X_t^\mu, \mu_t, \alpha_t) dt + \sigma dW_t + \sigma_0 dW_t^0, \quad X_0^\mu \sim \mu_{\text{init}}, \quad (1)$$

for a candidate measure flow $\mu = (\mu_t)_{t \in [0, T]}$, and the agent maximises

$$J^\mu(\alpha) = \mathbb{E} \left[\int_0^T f(t, X_t^\mu, \mu_t, \alpha_t) dt + g(X_T^\mu, \mu_T) \right]. \quad (2)$$

Assumption 2.1 (Lipschitz drift). *There exists $L_b > 0$ such that for all $t, x, x', \mu, \mu', \alpha, \alpha'$,*

$$\|b(t, x, \mu, \alpha) - b(t, x', \mu', \alpha')\| \leq L_b(\|x - x'\| + W_2(\mu, \mu') + \|\alpha - \alpha'\|).$$

Assumption 2.2 (Lipschitz rewards). *f and g are Lipschitz continuous in all arguments, with constants $L_f, L_g > 0$.*

Assumption 2.3 (Regularity and Lions differentiability). *b, f, g are C^1 in x and Lions-differentiable in μ (in the sense of [Carmona and Delarue, 2018, Ch. 5](#)), with bounded, Lipschitz Lions derivatives.*

Assumption 2.4 (Bounded volatility). *$\sigma > 0$ is constant, and $\sigma_0 \in \mathbb{R}^{d \times d_0}$ satisfies $\|\sigma_0\|_F \leq M_\sigma < \infty$.*

Assumption 2.5 (Uniform Hamiltonian concavity). *$\alpha \mapsto H(t, x, \mu, p, \alpha) := p \cdot b(t, x, \mu, \alpha) + f(t, x, \mu, \alpha)$ is uniformly λ_H -concave for some $\lambda_H > 0$.*

Assumption 2.6 (Initial condition). *$\mu_{\text{init}} \in \mathcal{P}_2(\mathbb{R}^d)$ has finite fourth moment.*

Under ?? 2.1–2.3?? 2.5, the Stochastic Maximum Principle characterises the optimal control for a fixed μ : there is a unique solution $(X^\mu, p^\mu, q^\mu, r^\mu, \alpha^\mu)$ to

$$\begin{aligned} dX_t^\mu &= b(t, X_t^\mu, \mu_t, \alpha_t^\mu) dt + \sigma dW_t + \sigma_0 dW_t^0, & X_0^\mu &\sim \mu_{\text{init}}, \\ -dp_t^\mu &= \partial_x H(t, X_t^\mu, \mu_t, p_t^\mu, \alpha_t^\mu) dt - q_t^\mu dW_t - r_t^\mu dW_t^0, & p_T^\mu &= \partial_x g(X_T^\mu, \mu_T), \\ \alpha_t^\mu &= \arg \max_{\alpha} H(t, X_t^\mu, \mu_t, p_t^\mu, \alpha), \end{aligned} \quad (3)$$

following standard results for Lipschitz McKean–Vlasov FBSDEs ([Carmona and Delarue, 2018, Vol. I, Ch. 4](#)). This defines the fixed-point map

$$\Phi(\mu)_t := \mathcal{L}(X_t^\mu \mid \mathcal{F}_t^{W^0}), \quad t \in [0, T], \quad (4)$$

on the space of adapted measure flows $\mathcal{M} := D([0, T]; \mathcal{P}_2(\mathbb{R}^d))$ (the Skorokhod space of càdlàg measure-valued processes adapted to $(\mathcal{F}_t^{W^0})$). A mean-field equilibrium is a fixed point $\Phi(\mu^*) = \mu^*$.

For $\lambda > 0$, define the weighted norm on $\mathcal{M}$,

$$\|\mu\|_{\lambda,T} := \sup_{t \in [0,T]} e^{-\lambda t} \mathbb{E}[W_2(\mu_t, \delta_0)^2]^{1/2},$$

and the corresponding distance $\|\mu^1 - \mu^2\|_{\lambda,T}$. This weighted norm is metrically equivalent to the standard norm of [Carmona and Delarue \(2018\)](#) but absorbs the Gronwall growth rate more conveniently for extracting an explicit contraction factor.

3 Stability Estimate for the Fixed-Point Map

The heart of the argument is a quantitative Lipschitz-type estimate for Φ .

3.1 Synchronous coupling

Given two candidate measure flows $\mu^1, \mu^2 \in \mathcal{M}$, we couple the two systems (3) on a common probability space, using the *same* realisations of the idiosyncratic noise W and, crucially, the same common noise W^0 . Since the Wasserstein distance between two laws is bounded above by the L^2 -distance between any coupling,

$$\mathbb{E}[W_2(\Phi(\mu^1)_t, \Phi(\mu^2)_t)^2] \leq \mathbb{E}[|X_t^1 - X_t^2|^2]. \quad (5)$$

3.2 The difference process

Write $\Delta X_t := X_t^1 - X_t^2$. Under synchronous coupling with identical, constant diffusion coefficients, the stochastic terms cancel *pathwise*:

$$d(\Delta X_t) = [b(t, X_t^1, \mu_t^1, \alpha_t^1) - b(t, X_t^2, \mu_t^2, \alpha_t^2)] dt,$$

so that Itô's formula applied to $|\Delta X_t|^2$ has no quadratic-variation correction:

$$d|\Delta X_t|^2 = 2\Delta X_t \cdot [b(t, X_t^1, \mu_t^1, \alpha_t^1) - b(t, X_t^2, \mu_t^2, \alpha_t^2)] dt. \quad (6)$$

Using Assumption 2.1 and Young's inequality,

$$d|\Delta X_t|^2 \leq [(2L_b + L_b^2)|\Delta X_t|^2 + L_b W_2(\mu_t^1, \mu_t^2)^2 + L_b |\alpha_t^1 - \alpha_t^2|^2] dt. \quad (7)$$

3.3 Control difference via the SMP

By the implicit function theorem applied to the optimality condition $\partial_\alpha H = 0$, uniform concavity (Assumption 2.5) gives a constant $C_\alpha > 0$ such that

$$|\alpha_t^1 - \alpha_t^2| \leq C_\alpha (|X_t^1 - X_t^2| + |p_t^1 - p_t^2| + W_2(\mu_t^1, \mu_t^2)). \quad (8)$$

The Lipschitz dependence on the measure argument follows from the Lions differentiability in Assumption 2.3 together with [Carmona and Delarue \(2018, Prop. 5.35\)](#).

3.4 Adjoint process estimate

Differencing the adjoint BSDEs gives a linear BSDE for $\Delta p_t := p_t^1 - p_t^2$ with a mean-field coupling term controlled, via Assumption 2.3, by a constant L_μ bounding the relevant Lions-derivative operator norm. Standard BSDE estimates ([El Karoui et al., 1997](#); [Zhang, 2017](#)) yield

$$\begin{aligned} & \mathbb{E} \left[\sup_{t \in [0,T]} |\Delta p_t|^2 + \int_0^T (|\Delta q_t|^2 + |\Delta r_t|^2) dt \right] \\ & \leq C_{\text{BSDE}} \mathbb{E} \left[|\Delta X_T|^2 + \int_0^T (|\Delta X_t|^2 + |\alpha_t^1 - \alpha_t^2|^2 + W_2(\mu_t^1, \mu_t^2)^2) dt \right], \end{aligned} \quad (9)$$

where C_{BSDE} depends on the Lipschitz bounds, T , and – crucially, through the martingale representation terms driven by W^0 – on $\|\sigma_0\|_F$.

3.5 Closing the estimate

Substituting (8) into (7) and (9) and setting $y_t := \mathbb{E}[|\Delta X_t|^2] + \mathbb{E}[|\Delta p_t|^2]$, algebraic manipulation (given in full in Appendix A) yields the closed differential inequality

$$\frac{d}{dt} y_t \leq C_1 y_t + C_2 \mathbb{E}[W_2(\mu_t^1, \mu_t^2)^2], \quad (10)$$

with

$$C_1 = 2L_b^2 + 2\|\sigma_0\|_F^2 + C_{\text{reg}}, \quad C_2 = L_b + C_\alpha L_b + C_{\text{BSDE}} L_\mu, \quad (11)$$

where C_{reg} collects the regularity constants from the Lions derivatives and BSDE bounds. By Gronwall's inequality and (5), this gives the main stability estimate:

Theorem 3.1 (Wasserstein Stability). *Under Assumptions 2.1–2.6, for all $\mu^1, \mu^2 \in \mathcal{M}$ and $t \in [0, T]$,*

$$\mathbb{E}[W_2(\Phi(\mu^1)_t, \Phi(\mu^2)_t)^2] \leq C_2 \int_0^t e^{C_1(t-s)} \mathbb{E}[W_2(\mu_s^1, \mu_s^2)^2] ds, \quad (12)$$

with C_1, C_2 as in (11).

Remark 3.2 (Expectation, not pathwise). The estimate (12) holds for the *expected* squared Wasserstein distance. The first-order stochastic terms driven by W^0 cancel in expectation (via the martingale property) but not pathwise; this is why the contraction below is stated in a weighted expectation norm rather than almost surely.

4 Main Theorem: Contraction, Existence, Uniqueness, and ε -Nash

Theorem 4.1 (Quantitative Well-Posedness). *Under Assumptions 2.1–2.6, for any $\lambda > \frac{C_1}{2} + \frac{C_2}{2}$, the map Φ is a strict contraction on the complete metric space $(\mathcal{M}, \|\cdot\|_{\lambda, T})$:*

$$\|\Phi(\mu^1) - \Phi(\mu^2)\|_{\lambda, T} \leq \rho \|\mu^1 - \mu^2\|_{\lambda, T}, \quad \rho = \sqrt{\frac{C_2}{2\lambda - C_1}} < 1. \quad (13)$$

Consequently, there exists a unique fixed point $\mu^* = \Phi(\mu^*) \in \mathcal{M}$, and the corresponding processes $(X^*, p^*, q^*, r^*, \alpha^*)$ solving (3) with $\mu = \mu^*$ constitute the unique solution to the common-noise MFG equilibrium system. Moreover, if all N players in the associated finite-player game adopt the strategy α^* , the resulting strategy profile is an ε_N -Nash equilibrium with $\varepsilon_N = O(N^{-1/2})$.

Proof sketch. The contraction bound (13) follows by taking the weighted supremum of (12) and bounding the resulting convolution integral, exactly as in the derivation of (11); see Appendix B for the full computation. Existence and uniqueness of the fixed point then follow from the Banach fixed-point theorem, since $(\mathcal{M}, \|\cdot\|_{\lambda, T})$ is complete (inherited from completeness of $\mathcal{P}_2(\mathbb{R}^d)$ and standard properties of Skorokhod space; Billingsley, 1999). The ε -Nash property follows from conditional propagation of chaos (Carmona and Delarue, 2018, Vol. II, Thm. 6.16), applied to bound the distance between the finite-player empirical measure and μ^* , combined with the Lipschitz stability of the FBSDE system established above. $\square$

5 Discussion of the Contraction Constant

The constant $C_1 = 2L_b^2 + 2\|\sigma_0\|_F^2 + C_{\text{reg}}$ in (11) admits a clean decomposition:

- $2L_b^2$: the standard Lipschitz contribution from the drift, present even without common noise;

- $2\|\sigma_0\|_F^2$: the effect of the common noise, entering not through the (cancelling) first-order stochastic term but through the dependence of the BSDE constant C_{BSDE} on $\|\sigma_0\|_F$ via the martingale representation for r_t ;
- C_{reg} : regularity constants from the Lions derivatives of b, f, g and the adjoint BSDE bounds. This term is fully explicit for specific models (e.g., the linear-quadratic case of Section 6) and otherwise bounded in terms of Lipschitz constants of the Lions derivatives.

Limitations. The estimate relies on absolute continuity and bounded densities for the stability of optimal transport maps (Assumption 2.3). In particle-method implementations, where the empirical measure is discrete, this assumption requires regularisation (e.g., entropic smoothing); the error introduced by such regularisation is not quantified here. The contraction constant ρ also grows with the horizon T through C_1, C_2 ; for long horizons this may require choosing a large λ , and under weaker structural conditions the contraction established here may only hold locally in time.

6 Illustration: The Linear-Quadratic Benchmark

Example 6.1 (Canonical LQ-MFG).

$$\begin{aligned} \text{State: } dX_t &= [\kappa(\bar{\mu}_t - X_t) + \alpha_t] dt + \sigma dW_t + \sigma_0 dW_t^0, \\ \text{Running cost: } f(t, x, \mu, \alpha) &= \frac{c_1}{2}x^2 + \frac{\lambda}{2}\alpha^2, \\ \text{Terminal cost: } g(x, \mu) &= \frac{c_2}{2}(x - \bar{\mu})^2, \end{aligned}$$

with $\bar{\mu}_t = \int x \mu_t(dx)$, $\kappa, \lambda, c_1, c_2 > 0$.

For this model, every constant in Theorems 3.1 and 4.1 is explicit in terms of $\kappa, \lambda, c_1, c_2, \sigma, \sigma_0$. The value function is $V(t, x, \mu) = -\frac{1}{2}A_t x^2 - B_t x \bar{\mu} - \frac{1}{2}C_t \bar{\mu}^2 - D_t x - E_t \bar{\mu} - F_t$, with $(A_t, \dots, F_t)$ solving a coupled Riccati system derived from the master equation for this model. With baseline parameters $\kappa = 0.5$, $\lambda = 1$, $c_1 = 2$, $c_2 = 5$, $\sigma = 0.2$, $\sigma_0 = 0.3$, $T = 1$, the Riccati system is well-posed on $[0, T]$ and yields an explicit contraction constant, verified numerically by two independent ODE solvers (LSODA and Radau). The full derivation of this Riccati system, including a re-derivation identifying and resolving two discrepancies against an earlier working version (a sign difference in the quadratic coefficient and missing mean-field coupling terms), is provided in a companion technical note.

7 Conclusion

We have established a quantitative, parameter-dependent contraction estimate for the fixed-point map characterising equilibria of common-noise mean-field games, extending the qualitative well-posedness theory of Carmona and Delarue (2018) and Ahuja et al. (2019) with an explicit rate suitable for algorithmic use. Natural directions for future work include extending the explicit-constant analysis beyond the linear-quadratic case, quantifying the regularisation error for discrete (particle-based) approximations, and using the contraction rate derived here as a design criterion for convergent numerical schemes – a direction pursued in a companion paper on numerical approximation of the coupled FBSDE–Fokker–Planck system.

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A Algebraic Details: Closing the Gronwall Inequality

This appendix gives the derivation of (10) from (7) and (9), together with the control bound (8).

Step 1: bound the control difference in the X -inequality. By (8) and the elementary inequality $(a + b + c)^2 \leq 3(a^2 + b^2 + c^2)$,

$$|\alpha_t^1 - \alpha_t^2|^2 \leq 3C_\alpha^2(|\Delta X_t|^2 + |\Delta p_t|^2 + W_2(\mu_t^1, \mu_t^2)^2).$$

Substituting into (7),

$$d|\Delta X_t|^2 \leq \left[(2L_b + L_b^2 + 3L_b C_\alpha^2) |\Delta X_t|^2 + 3L_b C_\alpha^2 |\Delta p_t|^2 + (L_b + 3L_b C_\alpha^2) W_2(\mu_t^1, \mu_t^2)^2 \right] dt. \quad (14)$$

Step 2: bound the control difference in the BSDE estimate. Applying (9) on the sub-interval $[0, t]$ (valid since $t \leq T$ is arbitrary and the same BSDE comparison argument applies verbatim on any sub-interval), and substituting the same bound on $|\alpha_s^1 - \alpha_s^2|^2$:

$$\begin{aligned} \mathbb{E} \left[\sup_{s \leq t} |\Delta p_s|^2 \right] &\leq C_{\text{BSDE}} \mathbb{E}[|\Delta X_t|^2] + C_{\text{BSDE}}(1 + 3C_\alpha^2) \int_0^t \mathbb{E}[|\Delta X_s|^2] ds \\ &\quad + 3C_{\text{BSDE}} C_\alpha^2 \int_0^t \mathbb{E}[|\Delta p_s|^2] ds + C_{\text{BSDE}}(1 + 3C_\alpha^2) \int_0^t \mathbb{E}[W_2(\mu_s^1, \mu_s^2)^2] ds. \end{aligned} \quad (15)$$

Step 3: a smallness condition closes the estimate. The term $3C_{\text{BSDE}} C_\alpha^2 \int_0^t \mathbb{E}[|\Delta p_s|^2] ds$ on the right-hand side of (15) involves the same quantity bounded on the left. We assume, as part of the standing regularity hypotheses (Assumption 2.3), that the model parameters satisfy

$$3C_{\text{BSDE}} C_\alpha^2 < 1; \quad (16)$$

this is a mild restriction on the size of C_α (control sensitivity) relative to C_{BSDE} (adjoint-process sensitivity), both of which are themselves determined by the Lipschitz and concavity constants of the model, and can always be arranged by working on a sufficiently short time interval or under a sufficiently strong concavity constant λ_H in Assumption 2.5. Under (16), a standard integral form of Gronwall's lemma applied to (15) (treating $\mathbb{E}[\sup_{s \leq t} |\Delta p_s|^2]$ as the unknown and absorbing the self-referential term via (16)) yields

$$\mathbb{E}[|\Delta p_t|^2] \leq \frac{C_{\text{BSDE}}}{1 - 3C_{\text{BSDE}} C_\alpha^2} \left(\mathbb{E}[|\Delta X_t|^2] + (1 + 3C_\alpha^2) \int_0^t \mathbb{E}[|\Delta X_s|^2 + W_2(\mu_s^1, \mu_s^2)^2] ds \right). \quad (17)$$

Step 4: combine. Adding (14) (in integrated form) and (17), and setting $y_t := \mathbb{E}[|\Delta X_t|^2] + \mathbb{E}[|\Delta p_t|^2]$, every term on the right is a constant multiple of either $\mathbb{E}[|\Delta X_s|^2]$, $\mathbb{E}[|\Delta p_s|^2]$ (hence of y_s), or $\mathbb{E}[W_2(\mu_s^1, \mu_s^2)^2]$, for $s \leq t$. Collecting terms and denoting $\tilde{C}_{\text{BSDE}} := C_{\text{BSDE}}/(1 - 3C_{\text{BSDE}} C_\alpha^2)$, one obtains, after routine algebra,

$$\frac{d}{dt} y_t \leq C_1 y_t + C_2 \mathbb{E}[W_2(\mu_t^1, \mu_t^2)^2],$$

where

$$C_1 = 2L_b + L_b^2 + 3L_b C_\alpha^2 + \tilde{C}_{\text{BSDE}}(1 + 3C_\alpha^2), \quad C_2 = L_b + 3L_b C_\alpha^2 + \tilde{C}_{\text{BSDE}}(1 + 3C_\alpha^2).$$

This is (10). The compact expression $C_1 = 2L_b^2 + 2\|\sigma_0\|_F^2 + C_{\text{reg}}$ quoted in (11) absorbs the constant $\tilde{C}_{\text{BSDE}}(1 + 3C_\alpha^2)$ and the $3L_b C_\alpha^2$ term into C_{reg} , and isolates the explicit $\|\sigma_0\|_F^2$ -dependence that enters through C_{BSDE} (and hence through $\tilde{C}_{\text{BSDE}}$) via the martingale representation for r_t in the adjoint BSDE, as discussed in Section 5.

Remark A.1 (On the smallness condition (16)). This condition does not appear explicitly in the informal derivation sketched in the main text; making it explicit here is, in the authors' view, necessary for a fully rigorous closure of the Gronwall argument, and we flag it as such rather than leaving it implicit. It is a standing structural assumption on the model (via $C_\alpha, C_{\text{BSDE}}$), not an additional constraint imposed after the fact; in the linear-quadratic case of Section 6 it can be checked directly from the explicit Riccati coefficients.

B Proof of the Contraction Bound

This appendix derives (13) from (12). Multiplying both sides of (12) by $e^{-2\lambda t}$ and taking the supremum over $t \in [0, T]$:

$$\begin{aligned} \|\Phi(\mu^1) - \Phi(\mu^2)\|_{\lambda, T}^2 &= \sup_{t \leq T} e^{-2\lambda t} \mathbb{E}[W_2(\Phi(\mu^1)_t, \Phi(\mu^2)_t)^2] \\ &\leq C_2 \sup_{t \leq T} e^{-2\lambda t} \int_0^t e^{C_1(t-s)} \mathbb{E}[W_2(\mu_s^1, \mu_s^2)^2] ds. \end{aligned}$$

Writing the integrand as $e^{-2\lambda(t-s)} e^{(C_1-2\lambda)s} \cdot e^{-2\lambda s} \mathbb{E}[W_2(\mu_s^1, \mu_s^2)^2]$ and using $e^{-2\lambda s} \mathbb{E}[W_2(\mu_s^1, \mu_s^2)^2] \leq \|\mu^1 - \mu^2\|_{\lambda, T}^2$ for every $s \leq T$, together with the change of variable $u = t - s$:

$$\|\Phi(\mu^1) - \Phi(\mu^2)\|_{\lambda, T}^2 \leq C_2 \left(\sup_{t \leq T} \int_0^t e^{-(2\lambda - C_1)u} du \right) \|\mu^1 - \mu^2\|_{\lambda, T}^2.$$

For $\lambda > C_1/2$, the integral is increasing in t and bounded by its value at $t = T$:

$$I(\lambda) := C_2 \int_0^T e^{-(2\lambda - C_1)u} du = \frac{C_2}{2\lambda - C_1} (1 - e^{-(2\lambda - C_1)T}) \leq \frac{C_2}{2\lambda - C_1}.$$

Choosing $\lambda > \frac{C_1}{2} + \frac{C_2}{2}$ ensures $2\lambda - C_1 > C_2$, hence $I(\lambda) < 1$, and $\rho := \sqrt{I(\lambda)} = \sqrt{C_2/(2\lambda - C_1)} < 1$. This gives

$$\|\Phi(\mu^1) - \Phi(\mu^2)\|_{\lambda, T} \leq \rho \|\mu^1 - \mu^2\|_{\lambda, T},$$

which is (13). Since $(\mathcal{M}, \|\cdot\|_{\lambda, T})$ is a complete metric space (Section 2), the Banach fixed-point theorem yields a unique $\mu^* \in \mathcal{M}$ with $\Phi(\mu^*) = \mu^*$, completing the proof of existence and uniqueness in Theorem 4.1. $\square$